Laborator 10

1. Pentru un graf reprezentat prin liste de adiacentă, implementati algoritmii lui Kruskal si Prim pentru determinarea arborilor minimali de acoperire.

```
MST-KRUSKAL(G, w)
A = ∅
for each vertex v ∈ G.V
    MAKE-SET(v)
sort the edges of G.E into nondecreasing order by weight w
for each edge (u, v) ∈ G.E, taken in nondecreasing order by weight
    if FIND-SET(u) ≠ FIND-SET(v)
        A = A ∪ {(u, v)}
        UNION(u, v)
return A
```

```
MST-PRIM(G, w, r)
for each u ∈ G.V
    u.key = ∞
    u.π = NIL
r.key = 0
Q = G.V
while Q ≠ ∅
    u = EXTRACT-MIN(Q)
    for each v ∈ G.Adj[u]
        if v ∈ Q and w(u, v) < v.key
            v.π = u
            v.key = w(u, v)
```

```
MAKE-SET(x)
x.p = x
x.rank = 0
```

```
FIND-SET(x)
if x ≠ x.p
    x.p = FIND-SET(x.p)
return x.p
```

```
UNION(x, y)
LINK(FIND-SET(x), FIND-SET(y))
```

```
LINK(x, y)
if x.rank > y.rank
    y.p = x
else x.p = y
    if x.rank == y.rank
        y.rank = y.rank + 1
```

```
MAIN()
    LISTA_ADIACENTA  G[…]
    PRINT “Nr noduri= ”
    READ n
    INIT_LISTA(G, n)
    FOR i:=1 TO N
        INSERT_VECINI(G, n, i) //vezi lab9
    END FOR
    PRINT “Lista adiacenta: ”
    PRINT_LISTA(G, n)
    PRINT “Kruskal: \n ”
    MST-KRUSKAL(G, w) //continuare →
    PRINT “Prim:  \n ”
    MST-PRIM(G, w, r)
END MAIN
```

2. Pentru un graf reprezentat prin liste de adiacenta, implementati algorimii Bellman-Ford si Djikstra pentru determinarea cailor de cost minim.

```
BELLMAN-FORD(G, w, s)
INITIALIZE-SINGLE-SOURCE(G, s)
for i = 1 to |G.V| − 1
    for each edge (u, v) ∈ G.E
        RELAX(u, v, w)
for each edge (u, v) ∈ G.E
    if v.d > u.d + w(u, v)
        return FALSE
return TRUE
```

```
INITIALIZE-SINGLE-SOURCE(G, s)
for each vertex v ∈ G.V
    v.d = ∞
    v.π = NIL
s.d = 0
```

```
DIJKSTRA(G, w, s)
INITIALIZE-SINGLE-SOURCE(G, s)
S = ∅
Q = G.V
while Q ≠ ∅
    u = EXTRACT-MIN(Q)
    S = S ∪ {u}
    for each vertex v ∈ G.Adj[u]
        RELAX(u, v, w)
```

```
RELAX(u, v, w)
if v.d > u.d + w(u, v)
    v.d = u.d + w(u, v)
    v.π = u
```

```
MAIN()
    LISTA_ADIACENTA  G[…]
    PRINT “Nr noduri= ”
    READ n
    INIT_LISTA(G, n)
    FOR i:=1 TO N
        INSERT_VECINI(G, n, i) //vezi lab9
    END FOR
    PRINT “Lista adiacenta: ”
    PRINT_LISTA(G, n)
    PRINT “Nod pornire: ” //continuare →
    READ nd
    BELLMAN-FORD(G, w, nd)
    DIJKSTRA(G, w, nd)
END MAIN
```